

PropManage Pro

Real-Time Property Rental, Maintenance &
Amenity Management Platform

Detailed Project Report

Project Title	PropManage Pro – Property Management Platform
Technology Stack	React.js, Tailwind CSS, Vite, Recharts, Lucide React
Project Type	Web Application (Frontend)
Year	2025 – 2026
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1. Introduction

PropManage Pro is a modern, fully responsive web-based platform built to streamline the management of residential properties, apartment complexes, and gated communities. The platform bridges the communication gap between residents and property administrators by providing a single unified digital interface for reporting issues, booking shared amenities, and monitoring overall property health.

In today's fast-paced world, traditional paper-based or manual processes for handling maintenance complaints and amenity reservations are inefficient and prone to delays. PropManage Pro eliminates these inefficiencies by digitizing the entire workflow, ensuring transparency, accountability, and real-time visibility for all stakeholders.

The application has been developed entirely using modern frontend technologies with no backend dependency, making it lightweight, fast, and easy to deploy on any static hosting platform such as Vercel, Netlify, or GitHub Pages.

2. Objectives

The primary objectives of this project are:

- To develop a user-friendly platform for residents to report maintenance issues with priority levels.
- To enable online booking of shared amenities such as gym, swimming pool, parking, and club house.
- To provide administrators with a comprehensive panel to manage all requests and bookings.
- To implement role-based access control differentiating tenants from administrators.
- To present real-time statistics and visual analytics through interactive charts and dashboards.
- To create a fully responsive design compatible with mobile, tablet, and desktop devices.
- To deliver a production-quality UI with modern design principles including dark/light mode.

3. Technologies Used

Technology	Version	Purpose
React.js	18.x	Core UI library for building component-based interfaces
Vite	5.x	Fast build tool and development server
Tailwind CSS	3.x	Utility-first CSS framework for responsive styling
Recharts	2.x	Declarative chart library built on D3 for React

Lucide React	0.38x	Lightweight, customizable SVG icon library
Node.js & NPM	20.x	JavaScript runtime and package manager

4. System Architecture

PropManage Pro follows a single-page application (SPA) architecture. The entire application logic resides within a single React component file (App.jsx), which leverages React's built-in state management (useState, useEffect, useCallback) to handle all dynamic behaviour without the need for a backend server or external database.

Layer	Component	Description
Presentation	React Components	Functional components with JSX for UI rendering
State Management	React Hooks	useState, useEffect, useCallback for local state
Styling	Tailwind CSS	Utility classes applied inline via className prop
Data	In-Memory Arrays	Dummy JSON arrays simulate a real database
Charts	Recharts	SVG-based responsive chart components
Icons	Lucide React	Tree-shakeable SVG icon components
Routing	Conditional Render	Page switching via React state (no router needed)

5. Features & Modules

Authentication System

- Secure login and registration pages with form validation.
- Role-based access: Tenant and Admin roles supported.
- Show/hide password toggle and error handling.
- Simulated async loading state with spinner feedback.

Dashboard

- Hero section with platform-wide summary.
- 4 KPI stat cards: Total Requests, Pending, In Progress, Completed.
- Interactive Area Chart for monthly request trends.
- Pie Chart for request status distribution.
- Bar Chart for monthly amenity bookings.
- Recent activity feed showing latest maintenance requests.

Maintenance Management

- Create, edit, and delete maintenance requests.
- Fields: Issue Title, Description, Priority (Low/Medium/High).
- Status tracking: Pending, In Progress, Completed.
- Filter requests by status with instant UI update.
- One-click status cycling directly from the card.
- Hover-revealed action buttons for clean UI.

Amenity Booking System

- 4 amenities: Fitness Center, Swimming Pool, Parking, Club House.
- Each card shows availability, capacity, and operating hours.
- Booking modal with date, check-in, and check-out time.
- Duplicate slot detection prevents double booking.
- My Bookings tab with cancellation option.
- Toast notifications on successful booking.

Admin Panel

- Accessible only to users with Admin role.
- Full maintenance request table with inline status dropdown.
- Full bookings table with delete capability.
- Real-time search across all records.

- Status filter for maintenance requests.
- Summary stat cards at the top of the panel.

6. Module Details

6.1 User Roles

Role	Access	Capabilities
Tenant / Resident	All modules except Admin Panel	Submit & track maintenance, book amenities, view dashboard
Administrator	Full access including Admin Panel	All tenant capabilities + manage all requests, update statuses,

6.2 Maintenance Priority Levels

Priority	Color Indicator	Use Case
High	Red Badge	Urgent issues affecting safety (broken window, gas leak)
Medium	Amber Badge	Important but not urgent (AC not cooling, drain clog)
Low	Green Badge	Minor issues (flickering light, cosmetic damage)

6.3 Amenities Available

Amenity	Capacity	Hours	Status
Fitness Center	20 persons	6 AM – 10 PM	Available
Swimming Pool	30 persons	7 AM – 9 PM	Available
Parking Garage	50 vehicles	24/7	Unavailable
Club House	60 persons	8 AM – 11 PM	Available

7. Data Flow

Since PropManage Pro is a frontend-only application, data flow is managed entirely through React state. Below is the simplified flow:

Step	Action	Result
1	User logs in with credentials	Matched against in-memory USERS_DB array; user object stored in state
2	User submits maintenance request	New object added to maintenance state array; UI re-renders

3	Admin updates request status	Target object updated in maintenance array via map(); badge updates
4	User books an amenity	Duplicate check runs; if clear, booking object added to bookings state
5	Admin deletes a booking	Object filtered out of bookings array; table re-renders instantly
6	Dashboard loads	Derived stats computed from maintenance and bookings arrays in real-time

8. UI/UX Design Highlights

Dark / Light Mode	Users can toggle between dark and light themes. The dark class is applied to the root HTML element, and all components respond via Tailwind's dark: variant prefix.
Responsive Sidebar	The sidebar collapses on mobile screens and is revealed via a hamburger menu with a backdrop overlay. On desktop (lg breakpoint), it is always visible.
Toast Notification System	A custom hook (useToast) manages a queue of toast messages. Each toast auto-dismisses after 3.5 seconds with a slide-in animation.
Loading States	Async operations (login, form submit, booking) show a Loader2 spinner with disabled button state to prevent double submissions.
Empty States	All list/table views display a contextual empty state illustration and message when no data matches the current filter or search.
Gradient Hero Section	The dashboard features a full-width gradient hero banner with decorative blurred circle elements for visual depth.
Hover Animations	Cards reveal action buttons on hover using group / group-hover Tailwind classes with opacity transitions.
Color-Coded Badges	Priority and status badges use consistent color semantics: Red=High/Error, Amber=Pending/Warning, Green=Completed/Success, Blue=In Progress.

9. Future Enhancements

The current version of PropManage Pro is a frontend-only prototype. The following enhancements are planned for future releases:

- Backend integration using Node.js + Express.js with a PostgreSQL or MongoDB database.
- Real-time notifications using WebSockets or Firebase Cloud Messaging.
- Email and SMS alerts for maintenance status updates.
- Payment gateway integration for rent collection and amenity fees.
- Document management for lease agreements and invoices.
- Mobile application using React Native sharing the same component logic.
- AI-based maintenance priority prediction using historical data patterns.
- Multi-property support for property management companies.
- Tenant rating and feedback system for maintenance technicians.

- Calendar integration for amenity booking with Google Calendar sync.

10. Conclusion

PropManage Pro successfully demonstrates the power of modern frontend technologies in solving real-world property management challenges. By leveraging React.js functional components with hooks, Tailwind CSS utility-first styling, and Recharts data visualisation, the project delivers a production-quality user experience without any backend dependency.

The platform effectively addresses the needs of both residents and administrators through its intuitive interface, role-based access control, and comprehensive feature set. The responsive design ensures accessibility across all device sizes, while the dark/light mode toggle caters to user preferences.

This project serves as a strong foundation for a full-stack property management solution and demonstrates proficiency in modern React development practices, component architecture, state management, and UI/UX design principles.

11. References

- [1] **React.js Official Documentation** — <https://react.dev>
- [2] **Tailwind CSS Documentation** — <https://tailwindcss.com/docs>
- [3] **Vite Documentation** — <https://vitejs.dev>
- [4] **Recharts Documentation** — <https://recharts.org>
- [5] **Lucide React Icons** — <https://lucide.dev>
- [6] **Node.js Official Site** — <https://nodejs.org>
- [7] **Vercel Deployment Platform** — <https://vercel.com>
- [8] **MDN Web Docs – React Hooks** — <https://developer.mozilla.org>